MONISHWAR REDDY VARDIREDDY

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Portfolio: Click here

SKILLS

- **Programming:** Python, C, R, Latex.
- **Tools:** VS Code, Jupyter Notebook, Google Colab, Overleaf.
- **ML Libraries:** NumPy, Pandas, Matplotlib, Scikit-learn, Tensorflow.
- **ML Models:** Linear Regression, Logistic Regression, Decision Tree, Random Forest, Support Vector Machine (SVM), K-Means, K-Nearest Neighbors (KNN),
- **Feature Extraction:** Discrete Wavelet Transform (DWT), Singular Value Decomposition(SVD)

EDUCATION

Bachelor of Technology
(Computer Science Engineering) (2023-2027)
Woxsen University, Hyderabad, Telangana, India
(CGPA: 8.61)

Intermediate (10+2), (2021-2023)
Narayana Junior College, Hyderabad, Telangana, India, Percentage: 83.7%

SSC, (2021)
Sri Chaitanya Techno School, Hyderabad, Telangana, India, GPA: 9.2

CERTIFICATIONS

- Edge Computing (NPTEL)
- Google AI Professional Certificate
- Machine Learning with Python.
- Probability & Statistics for Machine Learning & Data Science.
- Python Programming Essentials.
- Crash Course on Python.
- AI For Everyone.

View the Course Certificates here

LANGUAGES

- English
- Telugu
- Hindi

PROFESSIONAL SUMMARY

B.Tech Computer Science student with research experience in data analysis, biomedical signal processing, and EEG-based systems. Strong background in statistical modeling, mathematical modeling, and Monte Carlo simulations for predictive analysis and system optimization. Experienced in myocardial infarction detection and biomedical signal classification using machine learning and deep learning techniques, including CNN-based architectures and feature extraction methods. Contributed to interdisciplinary research initiatives and academic leadership programs supporting technology-driven innovation.

EXPERIENCE

PRME | Responsible Work & Purposeful Careers – Member (May 2026 – Present)
Main outcomes: Sustainability Initiatives, Research Collaboration.

- Engaged in initiatives promoting responsible work, sustainability, and purpose-driven careers.

Woxsen University – Research Assistant (May 2025 – Present)
Main outcomes: Epidemic Modelling, SEIR, Monte Carlo, and Predictive Analytics.

- Researched SEIR epidemic models, Monte Carlo simulations, and predictive systems, reviewing 50+ peer-reviewed reference papers.
- Generated insights on disease spread and probabilistic behavior to strengthen research outputs.

Woxsen Agentic Lab – Research Assistant (Dec 2025 – Mar 2026)
Main outcomes: Blockchain Research, Decentralized Systems, Entanglement Protocol Design.

- Worked on blockchain entanglement and decentralized coordination to enable 1 control access.
- Analyzing gas economics, finality, and censorship resilience.
- Building simulation frameworks for secure multi-party systems.

Woxsen University – Research Intern (Feb 2024 – Jul 2025)
Main outcomes: ECG Signals, HMI, Research.

- Contributed to Brain–Computer Interface (BCI) research projects, conducted EEG analysis.
- Authored a systematic literature review (117 references) and a book chapter published with IGI Global.

HCL Tech – Biomedical Signal Processing Intern (Jun 2025 – Jul 2025)
Main outcomes: Cardiac Classification, ECG Processing, CNN Model, Deep Learning.

- Designed a lightweight CNN integrating DWT + SVD with complete preprocessing and feature fusion.
- achieving 98.35% sampling accuracy and 88.25% validated model accuracy (journal publication).

Tiaano – R&D Analyst Intern (Jul 2024 – Nov 2024)
Main outcomes: Experimental Research, Data Analysis, Insight Generation.

- Conducted dataset analysis using data from min 6 companies with research-driven performance evaluation.
- Delivered strategic insights adopted in implementation through detailed research reports.

Woxsen University – Data Analyst Intern (Oct 2023 – Apr 2024)
Main outcomes: Social Media Strategy, Engagement Metrics, Content Optimization.

- Developed social media strategies and analyzed LinkedIn engagement metrics.
- Enhanced university visibility and audience engagement through digital content optimization.

View the Internship Certificates here

RESEARCH & PUBLICATIONS

- Vardireddy, Monishwar, Pranjali Gajbhiye, and Hemachandran Kannan. "**Exploring Music Effects on Human Emotion Emerging Through EEG: A Systematic Literature Review.**" Journal of Cortex Explore 1.1 (2026): 60-79.

CONFERENCES

Analytical and Numerical Methods in Differential Equations (Yanenko 105)

- Presented my Research Work on **Analysis of Epidemic Models through Singularities**

2025 UNC Greensboro Virtual PDE Conference.

- Presented my Research Work on **Analysis of SEIR Model around the Singularities.**

Symmetry2024 Conference.

- Presented my paper on **Review of Solution Procedures of Certain Ecological Models.**

CHEMFLUX 11.0.

- Presented my Research on **Analysis on Certain Ecological Models Applied on Real Life.**

REFERENCES

Dr. Amlan Kantri Halder
Head of Mathematics, WoxsenUniversity, Hyderabad
Email: amlankanti.halder@woxsen.edu.in

Prof. PGL Leach
Honorary ResearchProfessor, Durban University of Technology, South Africa
Email: leachp@ukzn.ac.za

Dr. Pranjali Gajbhiye
Neuroscientist, Woxsen University, Hyderabad
Email: pranjali.gajbhiye@woxsen.edu.in

- Venkatanarayana Pappula, Karre Saisuvan Reddy, Abhishek Reddy, Monishwar Reddy Vardireddy. **Various Methods For Plastic Waste Pyrolysis To Fuels: A Review. Journal of Polymer and Composites.** 2025; 13(04):79-93.

- Vardireddy, Monishwar Reddy, and Pranjali Gajbhiye. "**Integrating EEG Analysis to Illuminate Consumer Behavior and Optimize Marketing Strategies.**" The Quantum AI Era of Neuromarketing. IGI Global Scientific Publishing, 2025. 157-172.

- **RAMANUJAN EQUATION: AN APPROACH THROUGH SYMMETRY (Accepted).**

- **DWT–SVD-based Feature Extraction with CNN for the Detection of Myocardial Infarction using ECG Signal (Accepted).**

ACHIEVEMENTS

- Forge Alumnus × Microsoft Code-a-thon - **Top 4th team.**
- Agentathon 2025 - **Guinness World Record Certificate.**
- Code with Kiro Hackathon - **Bonus Blog Prize Winner.**
- Udhgam 2025 DigiTech Hackathon - **3rd Prize Winner.**
- RMEPT (Mauritania) 2025 - **Certificate of Recognition**

KEY PROJECTS

CampusConnect **Nov 2025 - Nov 2025**

A comprehensive AI-powered platform that makes student life easier, smarter, and more fun. From AI moderation, auto-tagging, and summaries to a spooky dark UI, this is the ultimate campus community experience.

AI Risk Dashboard **Nov 2025 - Nov 2025**

AI Environment Dashboard is a smart, serverless web app that visualizes real-time environmental data such as temperature, humidity, and air quality. It uses Flask and Google Cloud Run, with AI-powered logic through Google AI Studio (Gemini API).

AI Interactive Dashboard **Nov 2025 - Nov 2025**

AI Tutor is an educational web application demonstrating the power of AI-assisted learning. Users paste code, problems, or any text content and receive instant, step-by-step explanations powered by Google's Gemini AI model.

-More about my projects on GitHub.